

Deep learning for pancreatic cancer detection: current challenges and future strategies



In *The Lancet Digital Health*, Kao-Lang Liu and colleagues¹ describe the applications of a convolutional neural network (CNN) in distinguishing CT images of pancreatic cancer tissue from non-cancerous pancreatic tissue.¹ Contrast-enhanced CT images of 370 patients with pancreatic cancer and 320 controls from a Taiwanese centre were manually segmented and CNN was trained to classify image patches as cancerous or non-cancerous. Performance of CNN was compared with radiology reports in the local test sets. Similar to results from previous studies on this topic,^{2,3} CNN achieved a remarkable performance, with accuracy of 0.986–0.989 in the local test dataset. CNN achieved higher sensitivity than that of radiologists in the local test sets (0.983 vs 0.929; $p=0.014$). The three tumours that were missed by the CNN were 1.1–1.2 cm in size, of which two were correctly classified by radiologists. Impressively, CNN was able to correctly classify 11 of the 12 tumours that were missed by radiologists, which were 1.0–3.3 cm in size.¹ These promising results show that CNN as a second reader can reduce misdiagnosis of pancreatic cancer and possibly lead to improved patient outcomes.

Whereas previous publications relied on data from a single institution,^{2,3} Liu and colleagues also externally validated data with publicly available datasets from the USA, with an accuracy of 0.832. Importantly, external datasets of individuals with normal pancreas and pancreatic cancer patients originated from different sources: 82 patients with healthy pancreas from the Cancer Imaging Archive dataset and 281 patients with pancreatic cancer from the Medical Segmentation Decathlon dataset from Memorial Sloan Kettering Cancer Center, New York, NY, USA. Ideally, all patients from the external data should be from the same institution to ensure that the classification task is not affected by technical differences between the two institutions. The authors showed that a significant positive correlation existed between tumour size and CNN performance on both the internal and external datasets, which suggested that the CNN detected actual differences in pancreatic imaging features, rather than technical differences.

These preliminary experiences in applying a CNN to detect pancreatic cancer show tremendous promise, but

more external validation with datasets from multiple institutions is needed. The rate-limiting step lies in the curation of large, high-quality, annotated datasets that are publicly available, which will allow researchers to compare the performance of CNN models.⁴ However, curation of imaging data is an expensive and labour-intensive process that requires expertise in imaging anatomy and pathology, which limits the ability for crowdsourcing.^{5,6} Additionally, logistical and ethical concerns exist regarding data ownership and the protection of patient privacy that might hinder the development and sharing of public imaging datasets.^{6,7} To streamline the process, the Early Detection Research Network of the National Cancer Institute is in the preliminary stages of organising a centralised repository of pancreatic cancer imaging data.

Liu and colleagues showed that a CNN was able to achieve similar or even superior performance compared with that of radiologists. The goal for pancreatic cancer detection will be identifying pancreatic cancer before the subtle visual changes are apparent to a radiologist. One of the challenges in achieving this goal is the paucity of training data with these early subtle pancreatic cancers, because average-risk patients are not routinely screened for pancreatic cancer. The solution might be found with surveillance programmes in which patients at high risk of developing pancreatic cancer, based on family history or germline mutation, undergo serial imaging studies.⁸ In relation to the patients who eventually developed pancreatic cancer, the imaging exams before the diagnosis was made by the radiologist (pre-diagnostic exams) can be used retrospectively to train the CNN to recognise subtle imaging features that might signal the presence of an early stage pancreatic cancer. Another potential approach is to harness the longitudinal clinical and imaging data in large private or nationalised health systems, which might be able to provide such pre-diagnostic exams for training and testing.

Finally, Liu and colleagues' study, as with previous studies on pancreatic cancer detection,^{2,3} has been presented as including artificial binary constructs: cancer versus normal. However, radiologists encounter various types of pancreatic tumours in clinical practice,

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and non-neoplastic conditions such as pancreatitis can simulate the appearance of pancreatic tumors.^{9,10} Additional training will be necessary to distinguish pancreatic cancer from these other pathologies to improve the clinical utility of the model.

Artificial intelligence will certainly revolutionise the practice of radiology within the next decade. The collaboration of radiologists, computer scientists, and cancer researchers and the sharing of domain expertise will be essential in powering this revolution.

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